ULADiff: Unmixing-Guided Learnable Abundance-Latent Diffusion for Hyperspectral Image Denoising

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***Abstract*—Hyperspectral image (HSI) denoising is a critical preprocessing step in remote sensing. Recently, the denoising diffusion probabilistic models (DDPMs) have emerged as powerful generative models. However, applying DDPM to the HSI denoising task remains challenging owing to the scarcity and acquisition difficulty of HSIs. Thus, how to effectively incorporate physical priors into the DDPM to improve denoising performance remains an underexplored issue. To this end, we propose an unmixing-guided learnable abundance-latent diffusion (ULADiff) for HSI denoising, which is a from-scratch, task-specific diffusion framework that incorporates physically interpretable priors and conditional information into the DDPM. ULADiff comprises three key components, including a spectral unmixing transformer (SUT) network, an abundance-based diffusion model, and a reconstruction module. Specifically, we employ a learnable block-based SUT module in a self-supervised manner to decompose noisy HSIs into the abundance maps and endmembers. The SUT module enables the diffusion model to operate in a lower-dimensional abundance domain that better captures the underlying structure of HSIs. Then, we incorporate the first eigenimage, the reconstructed image via singular value decomposition (SVD), as a physically meaningful condition to facilitate controllable generation. Furthermore, we propose a reconstruction module that enforces a spatial–spectral consistency prior by simultaneously imposing total variation (TV) regularization on the endmembers and a sparsity constraint on the abundance maps. This design preserves the intrinsic structures of the HSI and improves reconstruction quality. Comprehensive evaluations on synthetic and real-world datasets demonstrate that ULADiff outperforms state-of-the-art methods in both quantitative performance and visual fidelity.**

***Index Terms*—Denoising diffusion probabilistic model (DDPM), hyperspectral image (HSI) denoising, total variation (TV), transformer.**

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I. Introduction

HYPERSPECTRAL images (HSIs) capture spectral information across numerous wavelengths, enabling subtle material identification and supporting a variety of applications. Applications include remote sensing [1], [2], [3], [4], [5], classification [6], [7], unmixing [8], and object tracking [9], [10]. However, due to the adverse imaging conditions, HSIs are susceptible to multiple types of noise, such as Gaussian, stripe, and impulse noise [11]. The presence of noise not only degrades HSI quality but also adversely affects downstream tasks. Therefore, developing effective denoising methods is essential for both visual enhancement and downstream task reliability.

Over the past few decades, substantial efforts have been devoted to HSI denoising. In general, existing methods can be broadly categorized into three types: model-based, deep-learning, and hybrid methods.

Model-based methods [12], [13], [14], [15], [16] address HSI denoising by formulating it as an optimization task, where hand-crafted priors are incorporated to constrain the solution space. These priors, such as low-rankness [12], nonlocal self-similarity [13], and total variation (TV) [14], have proven effective in modeling the intrinsic structures of HSIs. As a result, model-based methods are well-grounded in mathematical theory and offer high interpretability. Nevertheless, they often rely on subjective physical assumptions about HSIs. In contrast, deep-learning methods [17], [18], [19], [20], [21], [22], [23], [24] leverage deep neural networks to automatically learn implicit priors from large-scale noisy-clean HSI pairs. Despite achieving remarkable performance, they suffer from limited generalization due to the scarcity of paired HSIs. Moreover, deep-learning methods operate in a "black-box" manner and lack theoretical interpretability. Hybrid methods [25], [26], [27], [28], [29] integrate the advantages of both paradigms, aiming to provide a denoising framework with high interpretability and strong learning capacity. However, these methods often require manual tuning to balance model-based and deep-learning components, which may reduce flexibility and increase design complexity.

Denoising diffusion probabilistic models (DDPMs) [44] have recently demonstrated remarkable performance in image restoration tasks. Nevertheless, extending DDPMs to HSIs remains challenging owing to the limited availability of

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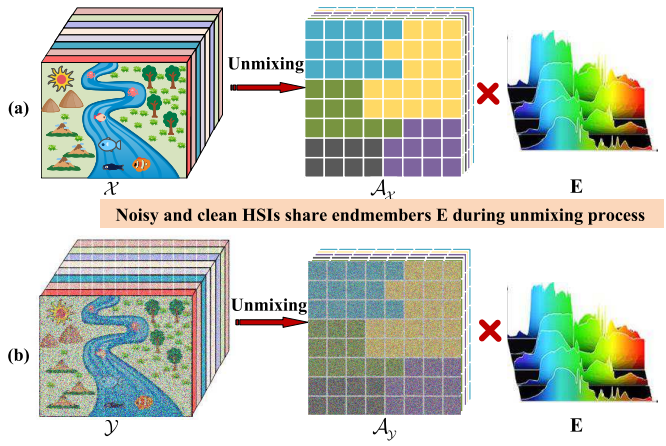


Fig. 1. Unmixing process on the clean HSI $\mathcal{X}$ and its noisy counterpart $\mathcal{Y}$. We assume that noisy HSI and clean HSI share the same endmembers E in the unmixing process, and noise mainly corrupts the abundance maps. (a) Clean image $\mathcal{X}$ unmixing. (b) Noisy counterpart $\mathcal{Y}$ unmixing.

training data and the high spectral dimensionality. Moreover, how to effectively incorporate physical priors and guidance into the DDPM remains a significant issue in HSI denoising.

To address the above issue, we propose an unmixing-guided learnable abundance-latent diffusion (ULADiff) for HSI denoising, which incorporates physically interpretable priors to guide the denoising process. Distinct from the existing methods, our ULADiff introduces two key innovations: 1) an explicit, physically interpretable guidance image to condition the diffusion model during training; and 2) a spatial–spectral consistency prior to enhance reconstruction quality. ULADiff is inspired by the physical insight that the HSI pixel can be modeled as a mixture of a few material components (endmembers) and their associated abundance maps. As shown in Fig. 1, we assume that clean and noisy HSIs share the same endmembers. Based on the insight, we propose ULADiff, which includes the spectral unmixing transformer (SUT) network [34], an abundance-based diffusion model, and a reconstruction module (see Fig. 2). First, we unmix HSIs into endmembers and abundance maps built upon a learnable block-based SUT module. The design offers a latent-space decomposition with clear physical interpretability. Second, we obtain the first eigenimage, which is the reconstructed image through singular value decomposition (SVD), exhibiting a higher signal-to-noise ratio than the others. This eigenimage serves as a clear, physically interpretable guidance image to condition the generative process. Finally, we design a reconstruction module that captures spatial–spectral consistency by applying a TV prior to the endmembers and a sparsity prior to the abundance maps. This improves the quality and fidelity of the reconstructed HSI. The primary contributions of this article can be outlined as follows.

1) We propose ULADiff, a latent diffusion framework for HSI denoising, which incorporates spectral unmixing into the DDPM as a physically interpretable prior, effectively bridging physical modeling and generative models.
2) We introduce a physically interpretable guidance image, namely the first eigenimage derived from the SVD, to improve interpretability and enhance the controllability of the diffusion process.
3) We propose a reconstruction module that enforces spatial–spectral consistency by applying a TV prior to the endmembers and a sparsity prior to the abundance maps. This enhances the quality of the reconstructed HSIs.

The remainder of this article is organized as follows. Section II reviews recent advances in HSI denoising and discusses the limitations of existing diffusion-based methods. Section III details the proposed ULADiff framework, covering the SUT module, the abundance-based diffusion model, and the reconstruction module. Section IV presents comprehensive evaluations on synthetic and real-world datasets. Finally, Section V summarizes the study and outlines potential directions for future work.

II. Related Work

In this section, we review recent advances in HSI denoising, which are model-based, deep-learning, and hybrid methods. Within deep-learning methods, we place particular emphasis on the diffusion-based methods.

A. Model-Based Methods

Model-based methods interpret HSI denoising as an ill-posed inverse problem and address it via regularization techniques. By incorporating hand-crafted priors about HSIs, these methods constrain the solution space and improve the stability of the inverse problem. Common priors include low-rankness [12], nonlocal self-similarity [13], and TV [14], [16]. Existing methods exploit one or more priors to tackle the ill-posed problem. For example, BM4D [15] applies 4-D collaborative filtering based on nonlocal means for HSI denoising. Enhanced3DTV [53] applies a sparsity prior to the spatial coefficients of each gradient map, facilitating edge preservation and spatial smoothness. LRTDTV [54] integrates Tucker decomposition with spatial–spectral TV to remove the mixed noise. R-PCA [55] performs HSI denoising by separating the observed data into the low-rank and sparse components. 3DCTV-RPCA [56] builds upon RPCA by employing 3-D TV constraints to better extract low-rank and locally smooth components. PWRCTV [57] incorporates an external prior from the panchromatic image and enforces an adaptive weighted TV on the representation coefficients to enhance denoising. BALMF [58] models each spectral band's noise with an asymmetric distribution to better capture the heavy-tailed characteristics of real HSI noise. While these methods offer strong interpretability, the priors are often subjective and fail to fully capture the image characteristics, leaving room for improvement. Moreover, they are often sensitive to parameter settings and require iterative optimization.

B. Deep-Learning Methods

Rather than resorting to hand-crafted priors, deep-learning methods directly learn image structures and details from large-scale noisy–clean paired images [17], [18]. For instance,

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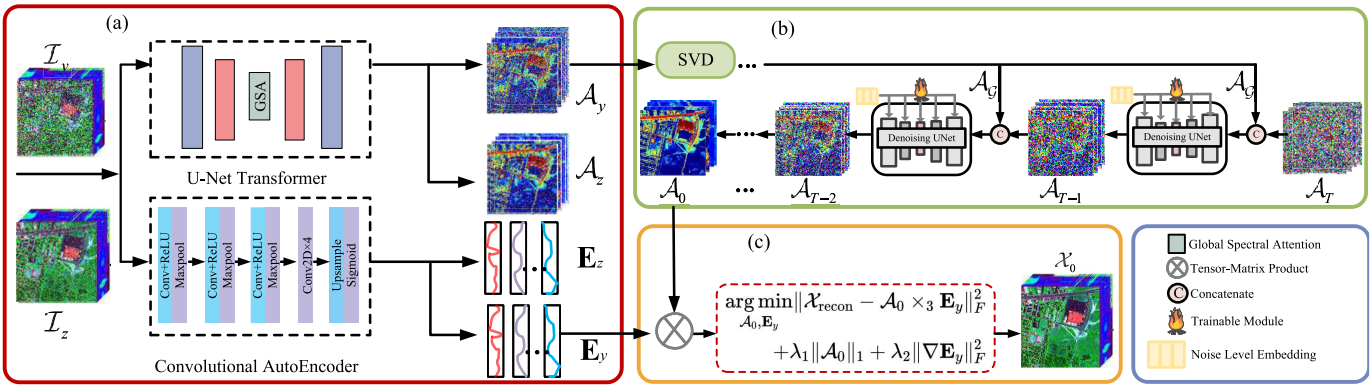


Fig. 2. Overall framework of ULADiff, which comprises three modules. (a) SUT network [34] decomposes the noisy HSI into the abundance maps and endmembers with physical interpretability. (b) First eigenimage, the reconstructed image by SVD, is used as effective guidance during training. (c) Reconstruction module enforces spatial–spectral consistency and improves the reconstruction quality.

Chang et al. [19] introduced multichannel 2-D filters to jointly capture the spatial and spectral structures. Building upon this, Yuan et al. [20] developed a deep neural network with the capability of multiscale feature extraction and achieved competitive denoising performance. Furthermore, Dosovitskiy et al. [21] introduced the vision transformers (ViTs), converting images into patch sequences to capture global relationships. These ViTs [21], [22], [23] demonstrate strong abilities to model global contextual interactions and nonlocal spatial dependencies in HSIs [24]. In summary, deep-learning methods achieve strong denoising performance when trained on large-scale paired HSIs. However, they often suffer from poor generalization ability due to the scarcity of labeled HSIs and the high acquisition cost. In addition, the lack of interpretability is an unavoidable issue in deep-learning methods.

C. Hybrid Methods

Hybrid methods leverage both the interpretability inherent in model-based methods and the representational ability of deep-learning methods. Representative paradigms include plug-and-play (PnP) [25], [26], deep image prior (DIP), and deep unfolding network (DUN) [27], [28]. PnP frameworks incorporate pretrained deep denoisers into an iterative optimization framework, introducing learned priors into the process. In contrast, DIP exploits the structure of an untrained neural network as an implicit prior. DUN unrolls optimization iterations into learnable network layers, offering interpretable architectures with end-to-end learning capabilities. In summary, hybrid models offer the dual advantages of interpretability and representation capabilities. Nevertheless, they require careful design to balance model-based and deep-learning components. Several methods exemplify this paradigm. For instance, Zhuang et al. [26] proposed HySuDeep, which leveraged spectral redundancy and employed FFDNet as the PnP denoiser. Xiong et al. [28] proposed SMDS-Net, which integrated a subspace-based sparse model with tensor regularizations for HSI denoising. Zhang et al. [29] introduced LRNet, embedding a PnP low-rank module implemented via matrix factorization into a convolutional network. Despite their promising performance, how to effectively integrate hand-crafted priors with deep representations remains a major problem in hybrid HSI restoration.

D. Diffusion Models for HSI Denoising

Recently, diffusion models originally developed for natural RGB images have been extended to HSI denoising. For instance, DDRM [30] addresses linear inverse problems in RGB restoration by applying SVD to the degradation matrix and conducting spectral-domain diffusion with a pretrained model. For HSI denoising, EGD-Net [31] performs conditional diffusion in an eigenimage subspace with a non-Gaussian forward process for mixed noise. However, its one-step sampling bypasses the multistep, iterative refinement process, which limits robustness against severe degradation. DDS2M [32] introduces a self-supervised framework incorporating a variational spatial–spectral module for HSI restoration. However, DDS2M fails to leverage prior knowledge of the datasets [33]. Moreover, its inference is computationally expensive, as each time step involves optimizing an untrained network. HIR-Diff [33] decomposes an HSI into the product of a reduced image and a coefficient matrix, thus reducing computational complexity in the restoration process. Similarly, Diff-Unmix [34] develops a self-supervised denoising method, integrating spectral unmixing with a fixed, pretrained diffusion model for HSI denoising. It employs a state-matching strategy to represent noisy abundance maps as intermediate states and performs reconstruction without joint regularization. In addition, the authors propose the SUT network to enhance the applicability of the decomposition process. Despite the demonstrated effectiveness of both HIR-Diff [33] and Diff-Unmix [34], their models rely heavily on off-the-shelf pretrained diffusion models[1] originally trained for HSI change detection tasks. Many parameters of these pretrained models are fixed and remain unchanged across different generation tasks, limiting their adaptability. To address the above issue, we propose a novel latent diffusion model for hyperspectral and multispectral image denoising. Unlike previous works that rely on fixed, pretrained diffusion models, our work is a physically interpretable and task-specific latent diffusion model trained from scratch for HSI denoising. Furthermore,

[1]https://github.com/wgcban/ddpm-cd/

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we explicitly incorporate physical priors into the denoising process and enforce spatial–spectral consistency via a reconstruction module.

III. ULADiff Framework

In this section, we first present the general formulation for HSI denoising and highlight the key difficulties of applying DDPM to this task. Then, we introduce the three modules of the ULADiff.

In this article, scalars are denoted by italic lowercase letters (e.g., x), vectors by bold lowercase letters (e.g., $\mathbf{x}$), matrices by bold uppercase letters (e.g., $\mathbf{X}$), and tensors by calligraphic uppercase letters (e.g., $\mathcal{X}$). Let $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ be a third-order tensor. Its mode-3 unfolding, the matrix $\mathbf{X}_{(3)} \in \mathbb{R}^{I_3 \times (I_1 I_2)}$, is constructed by rearranging the mode-3 fibers of $\mathcal{X}$ into columns of $\mathbf{X}_{(3)}$ in lexicographic order [35], [36]. The mode-3 product between a tensor $\mathcal{X}$ and a matrix $\mathbf{E} \in \mathbb{R}^{J_n \times I_3}$ is denoted by $\mathcal{Y} = \mathcal{X} \times_3 \mathbf{E}$, where $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times J_n}$. This operation is implemented via matrix multiplication on the unfolded tensor: $\mathbf{Y}_{(3)} = \mathbf{E}\mathbf{X}_{(3)}$. In addition, the element-wise ℓ_1 norm and Frobenius norm of $\mathcal{X}$ are defined, respectively, as $\|\mathcal{X}\|_1 = \sum_{i_1,i_2,i_3} |x_{i_1,i_2,i_3}|$ and $\|\mathcal{X}\|_F = \left(\sum_{i_1,i_2,i_3} |x_{i_1,i_2,i_3}|^2\right)^{1/2}$.

A. Problem Formulation

Let $\mathcal{X} \in \mathbb{R}^{M \times N \times p}$ represent a clean HSI, where M, N, and p correspond to the width, height, and number of bands, respectively. Under the general additive noise assumption, the observed degraded HSI can be expressed as follows:

$$\mathcal{Y} = \mathcal{X} + \mathcal{N} \tag{1}$$

where $\mathcal{Y} \in \mathbb{R}^{M \times N \times p}$ is the observed noisy HSI and $\mathcal{N} \in \mathbb{R}^{M \times N \times p}$ denotes Gaussian noise. Apparently, recovering $\mathcal{X}$ from $\mathcal{Y}$ is a highly ill-posed inverse problem. Although the prior-based methods are prone to being effective in addressing this problem [56], the exploration of more efficient integration of priors with generative networks remains underexplored.

Inspired by the broad success of the diffusion models, we leverage DDPM as a promising solution to HSI denoising. However, the high dimensionality and scarcity of HSIs present notable challenges for the training process, including high computational cost and the curse of dimensionality. Band-by-band HSI denoising using a pretrained three-band diffusion model neglects spectral correlation and requires multiple inferences, which is time-intensive; we provide a quantitative analysis later. In addition, another key issue lies in designing effective conditional information to guide the DDPM in generating accurate and natural outputs during training. This is partly because DDPM exhibits a weak correlation between the minimization of training loss and the quality of generated samples. In other words, a reduced loss value does not necessarily correspond to better generation performance. This inherent property makes it imperative to incorporate additional guidance during training to more effectively control the denoising process.

To address these issues, we propose ULADiff, a three-stage framework that integrates physical priors into generative models. Fig. 2 illustrates the overall architecture, which consists of the SUT network, the abundance-based diffusion model, and the reconstruction module. Below, we outline the pipeline and explain how each module mitigates the aforementioned issues.

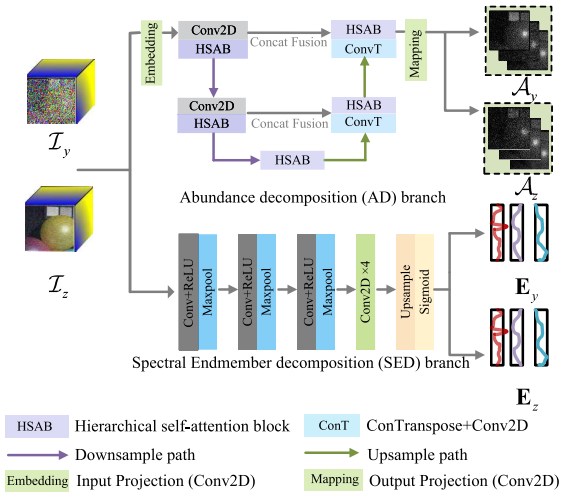


Fig. 3. Architecture of the SUT network, which comprises two branches, i.e., the AD branch and the SED branch. The details of the HSAB can be found in [34].

B. SUT Network

To address the issue, we adopt the SUT network proposed in [34]. The incorporation of the SUT module resolves two critical problems. **First**, it mitigates the curse of dimensionality and facilitates more efficient training of the DDPM. In particular, the architecture and reconstruction loss of the SUT network help preserve spectral correlations and mitigate information loss during the unmixing process. **Second**, by leveraging the SUT network, our model efficiently narrows the inference process to the latent abundance space and reduces inference time. This enhances both the efficiency and scalability of our approach for practical HSI denoising. In summary, the SUT module not only fulfills traditional unmixing objectives but also serves as a physically meaningful dimensionality-reduction method for the abundance-based diffusion model.

Based on the linear mixture model (LMM) [37], the clean HSI $\mathcal{X}$ is represented as a linear combination of abundance maps and endmembers. It can be formulated as follows:

$$\mathcal{X} = \mathcal{A} \times_3 \mathbf{E} \tag{2}$$

where $\mathcal{A} \in \mathbb{R}^{M \times N \times k}$ and $\mathbf{E} \in \mathbb{R}^{p \times k}$ denote the abundance maps and the spectral endmembers with k endmembers, respectively [38]. In spectral unmixing theory, the abundance maps illustrate the proportion of different spectral materials, referred to as endmembers, present within a mixed pixel. These abundance values quantify the contribution of each endmember to the overall spectral signature observed in the pixel. The endmember corresponds to the pure spectral signatures of individual materials within a scene, ideally without any mixture. Thus, endmembers tend to remain constant under various degradation conditions. Therefore, noise can be separated into the abundance maps (see Fig. 1).

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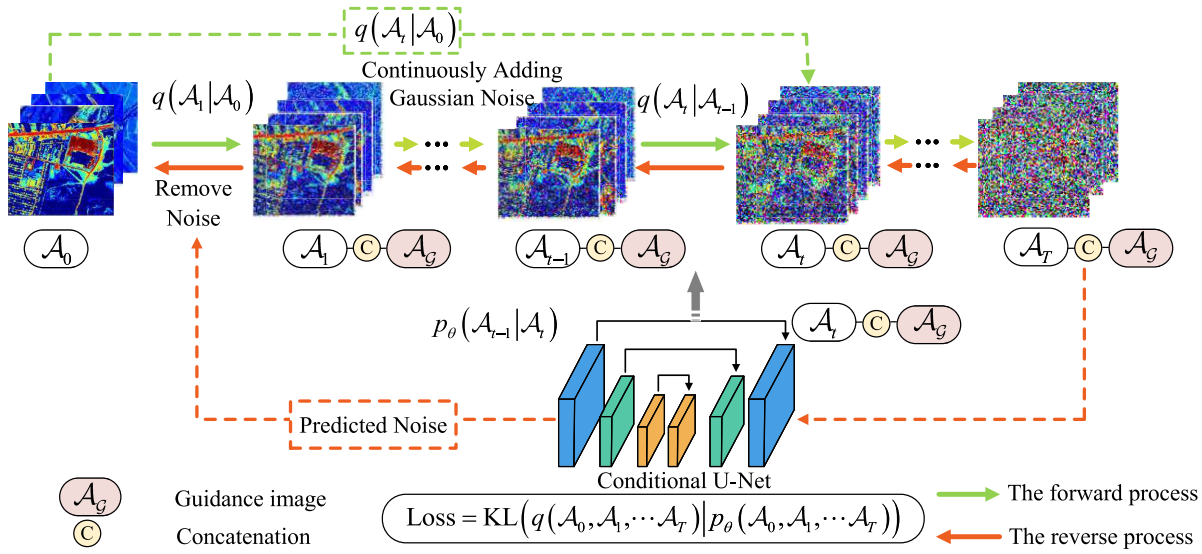


Fig. 4. Training procedure of the abundance-based diffusion model, where green lines denote the forward diffusion process and orange lines denote the reverse process. The network is trained to approximate the conditional distribution $p_\theta(\mathcal{A}_{t-1} \mid \mathcal{A}_t, \mathcal{A}_\mathcal{G})$ by predicting the noise. The training objective is the mean-squared error between the predicted noise and the true noise, which is equivalent to minimizing a Kullback–Leibler (KL) divergence term in the variational bound.

Under the above assumptions, we employ the SUT network [34] for the decomposition task. As illustrated in Fig. 3, this network adopts a dual-branch structure, i.e., the abundance decomposition (AD) branch and spectral endmember decomposition (SED) branch. The network integrates an autoencoder with global spectral attention (GSA) [24] to jointly extract spectral and spatial features. Specifically, the AD branch employs a U-shaped transformer, which incorporates hierarchical self-attention blocks (HSABs) in both downsampling and upsampling paths. It is worth noting that GSA serves as the core of each HSAB block, enabling effective modeling of long-range spectral dependencies. In contrast, the SED branch is a convolutional autoencoder. The SUT network takes as input a pair of noisy HSIs, namely the original noisy image $\mathcal{I}_y$ and its noise-variant counterpart $\mathcal{I}_z$. These inputs are jointly processed by the two branches to capture both spectral abundance maps and endmembers under different noise conditions. Specifically, the AD branch estimates the abundance maps $\mathcal{A}_y$ and $\mathcal{A}_z$, which describe the material distributions corresponding to $\mathcal{I}_y$ and $\mathcal{I}_z$. In parallel, the SED branch extracts low-dimensional representations from the input images, yielding the corresponding endmembers $\mathbf{E}_y$ and $\mathbf{E}_z$ [39]. To respect the physical constraints, the SED branch terminates with a sigmoid layer (see Fig. 3). The AD branch, on the other hand, concludes with a softmax layer to impose both the abundance sum-to-one constraint (ASC) and the abundance nonnegativity constraint (ANC). Following [34], the SUT loss function comprises three terms, namely a reconstruction loss $\mathcal{L}_{\text{rec}}$ to ensure fidelity of HSI reconstruction, an abundance consistency loss $\mathcal{L}_{\text{af}}$, and a spectral endmember consistency loss $\mathcal{L}_{\text{se}}$. For completeness, they are defined as follows:

$$\mathcal{L}_{\text{rec}} = \|\mathcal{I}_y - \mathcal{A}_y \times_3 \mathbf{E}_y\|_1 + \alpha_{\text{rec}} \|\mathcal{I}_z - \mathcal{A}_z \times_3 \mathbf{E}_z\|_1 \quad (3)$$

$$\mathcal{L}_{\text{af}} = \|\mathcal{A}_y - \mathcal{A}_z\|_1, \quad \mathcal{L}_{\text{se}} = \|\mathbf{E}_y - \mathbf{E}_z\|_1 \quad (4)$$

where α_{rec} is the hyperparameter. Further architectural details of the SUT network can be found in [34].

C. Abundance-Based Diffusion Model

To improve the DDPM training process and the generated sample quality, we condition the diffusion on a physically interpretable conditional image, i.e., the eigenimage obtained via SVD. This strategy is grounded in the observation that the reconstructed image using the dominant singular value captures the essential structural information and energy of the original image, while the smaller singular values primarily represent noise [40]. We refer to the reconstructed image as the first eigenimage, denoted as $\mathcal{A}_\mathcal{G}$. The introduction of $\mathcal{A}_\mathcal{G}$ effectively bridges spectral unmixing and the denoising task. As depicted in Fig. 4, the eigenimage is concatenated with the noisy input and serves as a conditioning signal during generation. Unlike previous approaches [33], [34], our method injects this physically interpretable guidance into the diffusion model during the training process. We adopt the architecture of the super-resolution via repeated refinement (SR3) framework [42]. The diffusion process consists of two Markov chains, namely a fixed forward process that progressively adds noise, and a learned reverse process that generates samples by denoising [2], [3], [40], [43].

1) *Forward Process*: Let $\mathcal{A}_0$ denote the initial abundance maps before noise is added, with data distribution represented as $q(\mathcal{A}_0)$. The forward process progressively adds Gaussian noise to the abundance maps over T time steps, producing a sequence of increasingly noisy abundance maps: $\mathcal{A}_0, \mathcal{A}_1, \ldots, \mathcal{A}_T$. After adding Gaussian noise T times, the abundance map $\mathcal{A}_0$ is transformed into $\mathcal{A}_T$. $\mathcal{A}_T$ approximately follows a standard Gaussian distribution, i.e., $\mathcal{A}_T \sim \mathcal{N}(0, \mathbf{I})$ [46]. During the process, the variance of the added Gaussian noise is controlled

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by the fixed parameter β_t, which is a predefined schedule that controls the noise variance. As the time step t increases, the data distribution $q(\mathcal{A}_t)$ gradually converges to a standard Gaussian distribution. Moreover, the forward process forms a Markov chain, in which each time step's distribution depends only on the previous time step. Then, the conditional distribution at time step t is given by the following equations:

$$q(\mathcal{A}_t \mid \mathcal{A}_{t-1}) = \mathcal{N}\left(\mathcal{A}_t;\ \sqrt{\bar{\alpha}_t}\,\mathcal{A}_0, (1-\bar{\alpha}_t)\,\mathbf{I}\right) \tag{5}$$

$$q\left(\mathcal{A}_{1:T} \mid \mathcal{A}_0\right) = \prod_{t=1}^{T} q\left(\mathcal{A}_t \mid \mathcal{A}_{t-1}\right). \tag{6}$$

With the reparameterization trick, the abundance map $\mathcal{A}_t$ at any time step t is determined by the initial map $\mathcal{A}_0$ and the noise level β_t. We define variables $\alpha_t = 1-\beta_t$, where α_t represents the fraction of the original signal preserved at each time step t, and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ denotes its cumulative product. With these definitions, $\mathcal{A}_t$ can be formulated as follows:

$$\mathcal{A}_t = \sqrt{\bar{\alpha}_t}\mathcal{A}_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon \tag{7}$$

where $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$ denotes the standard Gaussian noise. This leads to the marginal distribution

$$q\left(\mathcal{A}_t \mid \mathcal{A}_0\right) = \mathcal{N}\left(\mathcal{A}_t;\ \sqrt{\bar{\alpha}_t}\mathcal{A}_0, (1-\bar{\alpha}_t)\,\mathbf{I}\right). \tag{8}$$

2) *Reverse Process*: The reverse process is modeled as a learned Markov chain that progressively reconstructs $\mathcal{A}_0$ from the noisy signal $\mathcal{A}_T$, conditioned on the guidance $\mathcal{A}_\mathcal{G}$. Unlike the forward process with a fixed noise schedule, the reverse process cannot be explicitly formulated because it depends on the unknown data distribution. If the reverse distribution $q\left(\mathcal{A}_{t-1} \mid \mathcal{A}_t\right)$ were known, we could generate a realistic sample by progressively denoising from the terminal noisy state $\mathcal{A}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ [45]. However, since $q(\mathcal{A}_{t-1}|\mathcal{A}_t)$ depends on the entire data distribution and cannot be directly calculated, we employ a deep neural network (e.g., U-Net) to approximate the parameters of the distribution p_θ. Under the Markov assumption, (5) and (6), the reverse distribution p_θ can be expressed as follows:

$$p_\theta(\mathcal{A}_{0:T} \mid \mathcal{A}_\mathcal{G}) = p(\mathcal{A}_T) \prod_{t=1}^{T} p_\theta(\mathcal{A}_{t-1} \mid \mathcal{A}_t, \mathcal{A}_\mathcal{G}) \tag{9}$$

$$p_\theta(\mathcal{A}_{t-1} \mid \mathcal{A}_t, \mathcal{A}_\mathcal{G}) = \mathcal{N}\left(\mathcal{A}_{t-1};\ \mu_\theta(\mathcal{A}_t, \mathcal{A}_\mathcal{G}, t), \Sigma_\theta(\mathcal{A}_t, \mathcal{A}_\mathcal{G}, t)\right) \tag{10}$$

where μ_θ denotes the predicted mean of the reverse Gaussian distribution at time step t. $\sum_\theta(\mathcal{A}_t, \mathcal{A}_\mathcal{G}, t))$ is corresponding variance, both parameterized by a neural network. Based on the Bayes's theorem, the posterior distribution $q\left(\mathcal{A}_{t-1} \mid \mathcal{A}_t, \mathcal{A}_0\right)$ can be derived using β_t and $\tilde{\mu}_t\left(\mathcal{A}_t, \mathcal{A}_0\right)$. It is expressed as follows:

$$\tilde{\beta}_t = 1 \Big/ \left(\frac{\alpha_t}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}}\right) = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t} \cdot \beta_t \tag{11}$$

$$\tilde{\mu}_t\left(\mathcal{A}_t, \mathcal{A}_0\right) = \frac{\sqrt{\alpha_t}\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t}\mathcal{A}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathcal{A}_0 \tag{12}$$

where $\tilde{\mu}_t$ and $\tilde{\beta}_t$ represent the theoretical mean and variance of the true reverse posterior distribution, which are used to define the training objective. Based on (7), $\mathcal{A}_t$ is formulated as $\mathcal{A}_t = \left(\bar{\alpha}_t\right)^{1/2}\mathcal{A}_0 + \left(1-\bar{\alpha}_t\right)^{1/2}\varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$. Under the framework, a DL network is designed to predict the noise ε with inputs $\mathcal{A}_t$ and t. Then, we can obtain the expectation as follows:

$$\mu_\theta(\mathcal{A}_t, \mathcal{A}_\mathcal{G}, t) = \frac{1}{\sqrt{\alpha_t}}\left(\mathcal{A}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta(\mathcal{A}_t, \mathcal{A}_\mathcal{G}, t)\right). \tag{13}$$

Ultimately, we train the network by minimizing the following noise-prediction objective:

$$\mathcal{L}_{t-1} = \mathop{\mathbb{E}}_{\mathcal{A}_0, \mathcal{A}_\mathcal{G}, \varepsilon}\left[\left\|\varepsilon - \varepsilon_\theta\left(\sqrt{\bar{\alpha}_t}\mathcal{A}_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon, \mathcal{A}_\mathcal{G}, t\right)\right\|^2\right]. \tag{14}$$

During the reverse process, the abundance map at the previous time step can be sampled from a Gaussian distribution

$$\mathcal{A}_{t-1} = \mu_\theta(\mathcal{A}_t, \mathcal{A}_\mathcal{G}, t) + \sigma_t z, \quad z \sim \mathcal{N}(0, \mathbf{I}) \tag{15}$$

where $\sigma_t = \left(\beta_t\right)^{1/2}$ is the standard deviation at time step t.

Algorithm 1 Training of Abundance-Based Diffusion

Require: Noise schedule $\{\beta_t\}_{t=1}^{T}$ with $\alpha_t = 1-\beta_t$, $\bar{\alpha}_t = \prod_{s=1}^{t}\alpha_s$, conditional prior $\mathcal{A}_\mathcal{G}$, time steps T

Output: Pretrained denoising model ε_θ
while not converged **do**
  Sample clean abundance $\mathcal{A}_0$ and condition $\mathcal{A}_\mathcal{G}$ from $p(\mathcal{A}_0, \mathcal{A}_\mathcal{G})$
  Sample time step $t \sim$ Uniform$(\{1, 2, \ldots, T\})$
  Sample noise $\varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  Generate noisy Input: $\mathcal{A}_t = \sqrt{\bar{\alpha}_t}\mathcal{A}_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon$
  Optimize loss: $\mathcal{L} = \|\varepsilon - \varepsilon_\theta(\mathcal{A}_t, \mathcal{A}_\mathcal{G}, t)\|^2$ to update θ
end while

Algorithm 2 Sampling From Abundance-Based Diffusion

Require: Trained model ε_θ, condition $\mathcal{A}_\mathcal{G}$, time steps T

Output: The abundance maps $\mathcal{A}_0$
Initialize $\mathcal{A}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = T$ **down to** 1 **do**
  Sample $z \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  Predict noise: $\hat{\varepsilon}_\theta = \varepsilon_\theta(\mathcal{A}_t, \mathcal{A}_\mathcal{G}, t)$
  Update abundance map:

$$\mathcal{A}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathcal{A}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\hat{\varepsilon}_\theta\right) + \sigma_t z$$

end for
return $\mathcal{A}_0$

To approximate the distribution in the reverse process, we employ a hierarchical U-Net structure (see Fig. 5), following the existing DL-based methods [42], [47]. Specifically, the network consists of three components: 1) a noise-level embedding; 2) an encoder that downsamples features; and 3) a decoder that upsamples features. In particular, the noise level

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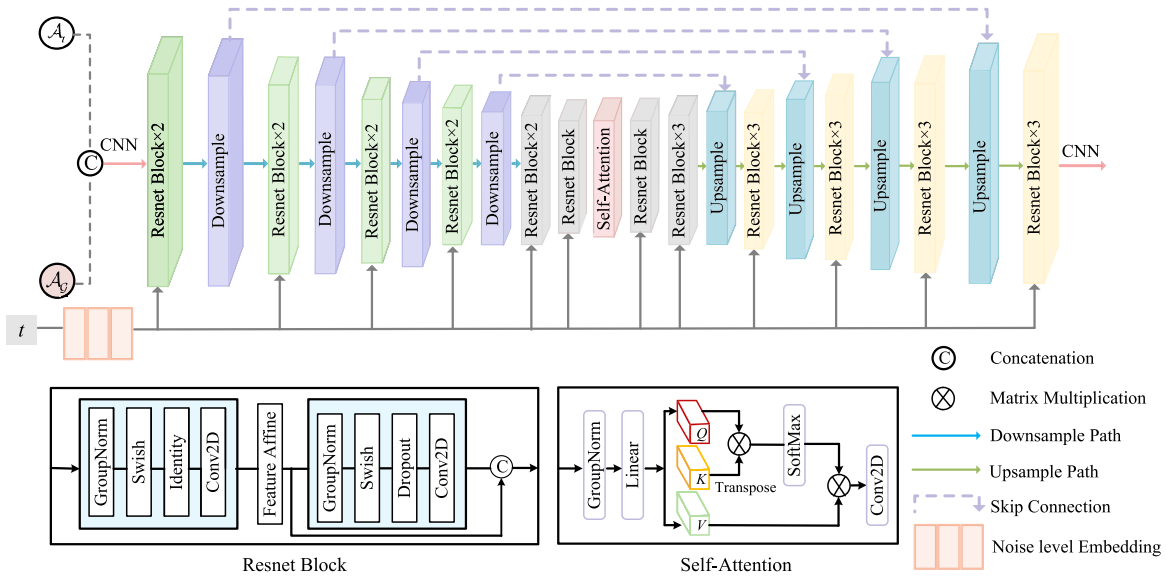


Fig. 5. Architecture of the U-Net backbone used in the reverse diffusion process.

embedding module maps the diffusion time steps using positional encoding [48], which is then processed by a two-layer MLP with Swish activations to condition subsequent layers. At the input, the encoder receives the concatenation of the noisy and conditional images. After that, multiple residual blocks and downsampling operations are applied to extract multiscale features while progressively reducing the spatial resolution. At the end of the encoder, a self-attention module is employed to capture global features and enhance long-range dependencies. The decoder reconstructs the image through the residual blocks and upsampling operators. Through skip connections that bridge the encoder and decoder, high-frequency information is transmitted effectively, enhancing the modeling of Gaussian-like high-frequency noise. Finally, a convolutional layer maps the features to the target image, completing the mapping from random noise to a high-quality image generation. We summarize the overall training and sampling procedures of the proposed abundance-based diffusion model in Algorithms 1 and 2, respectively.

D. Reconstruction Module

To improve the spatial–spectral consistency in the reconstructed HSI, we introduce a reconstruction module that jointly updates the endmembers and abundance maps. Endmember spectra generally vary smoothly along the spectral axis. Accordingly, we promote spectral smoothness by enforcing sparsity constraints on the gradient maps of the endmembers. Moreover, since each pixel typically comprises only a few materials, the corresponding abundance vectors are expected to be sparse. Consequently, we encourage sparsity by imposing an ℓ_1-norm regularizer on the abundance maps.

Building on the above priors, we initially reconstruct the HSI by mixing the diffusion-generated abundance maps with the estimated spectral endmembers. It can be expressed as follows:

$$\mathcal{X}_{\text{recon}} = \mathcal{A}_0 \times_3 \mathbf{E}_y \tag{16}$$

where $\mathcal{X}_{\text{recon}} \in \mathbb{R}^{M\times N\times p}$ is the reconstructed HSI, $\mathcal{A}_0 \in \mathbb{R}^{M\times N\times k}$ denotes the abundance maps from the DDPM, and $\mathbf{E}_y \in \mathbb{R}^{p\times k}$ is the endmembers estimated by the SUT module. To further enhance reconstruction quality, we formulate an optimization framework that jointly updates the generated abundance maps and endmembers. The reconstruction model is represented via

$$\arg\min_{\mathcal{A}_0,\mathbf{E}_y} \|\mathcal{X}_{\text{recon}} - \mathcal{A}_0 \times_3 \mathbf{E}_y\|_F^2 + \lambda_1\|\mathcal{A}_0\|_1 + \lambda_2\|\nabla\mathbf{E}_y\|_F^2 \tag{17}$$

where λ_1 and λ_2 are the regularization weights, and ∇ represents the gradient operator. Specifically, the endmembers could be updated by the following equation:

$$\mathbf{E}'_y = \arg\min_{\mathbf{E}_y} \left\|\mathcal{X}_{\text{recon}} - \mathcal{A}_0 \times {}_3\mathbf{E}_y\right\|_F^2 + \lambda_2 \left\|\nabla\mathbf{E}_y\right\|_F^2. \tag{18}$$

Then, the $\mathcal{A}_0$ can be updated alternatively

$$\mathcal{A}'_0 = \arg\min_{\mathcal{A}} \left\|\mathcal{X}_{\text{recon}} - \mathcal{A}_0 \times {}_3\mathbf{E}'_y\right\|_F^2 + \lambda_1\|\mathcal{A}_0\|_1. \tag{19}$$

Overall, the final HSI is reconstructed by the following equation:

$$\mathcal{X}_0 = \mathcal{A}'_0 \times_3 \mathbf{E}'_y. \tag{20}$$

IV. Experiments

In this section, we conduct a series of experiments to evaluate the feasibility and effectiveness of ULADiff from multiple perspectives.

A. Experimental Setting

1) Datasets: To comprehensively assess the performance of our method, we employ four publicly available HSI datasets for training and testing. Specifically, the datasets include the CAVE[2] dataset, KAIST[3] dataset, ICVL[4] dataset, and Harvard

[2]http://www1.cs.columbia.edu/CAVE/databases/multispectral/
[3]https://vclab.kaist.ac.kr/siggraphasia2017p1/kaistdataset.html
[4]https://icvl.cs.bgu.ac.il/hyperspectral/

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University[5] dataset for model training and testing. Key specifications of these datasets are as follows.

a) The CAVE dataset, also known as the Columbia Multispectral Image Database, comprises 32 images of size 512×512 pixels with 31 spectral bands, covering 400–700 nm at 10-nm sampling intervals [49].
b) The KAIST dataset, captured with the Point Gray Grasshopper 9.1 MP monochrome camera (GS3-U3-91S6M-C), contains 30 images of size 3376×2704 pixels and covers a spectral range of 420–720 nm [50].
c) The ICVL dataset, also known as the Interdisciplinary Computational Vision Laboratory dataset, is captured with the Specim PS Kappa DX4 hyperspectral camera. It comprises 201 images of size 1392×1300 pixels with 31 spectral bands covering 400–700 nm [51].
d) The Harvard University dataset, captured with a commercial hyperspectral camera (Nuance FX; CRI, Inc.), comprises 50 natural-scene images of size 1392×1040 pixels with 31 spectral bands covering 420–720 nm [52].

2) Evaluation Metrics: To quantitatively evaluate denoising performance across methods, we employ three commonly used evaluation metrics, i.e., peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), and erreur relative globale adimensionnelle de synthése (ERGAS). For PSNR and SSIM, we report the mean values across all spectral bands. The optimal values for PSNR, SSIM, and ERGAS are $+\infty$, 1, and 0, respectively. The definitions of PSNR, SSIM, and ERGAS are as follows:

$$\text{PSNR} = \frac{1}{B}\sum_{i=1}^{B}\text{PSNR}_i \quad \text{SSIM} = \frac{1}{B}\sum_{i=1}^{B}\text{SSIM}_i$$

$$\text{ERGAS} = 100 \cdot \sqrt{\frac{1}{B}\sum\nolimits_{i=1}^{B}\frac{\text{mse}(\text{ref}_i, \text{res}_i)}{\mu(\text{ref}_i)^2}}$$

where PSNR_i and SSIM_i are given as follows:

$$\text{PSNR}_i = 10 \cdot \log_{10}\left(\frac{\text{MAX}^2}{\text{mse}(\text{ref}_i, \text{res}_i)}\right)$$

$$\text{SSIM}_i = \frac{(2\mu_{\text{ref}_i}\mu_{\text{res}_i} + C_1)(2\sigma_{\text{ref}_i,\text{res}_i} + C_2)}{(\mu_{\text{ref}_i}^2 + \mu_{\text{res}_i}^2 + C_1)(\sigma_{\text{ref}_i}^2 + \sigma_{\text{res}_i}^2 + C_2)}.$$

Here, B is the number of spectral bands. ref_i and res_i denote the ith spectral band of the reference and reconstructed HSI, respectively. μ_{ref_i} and μ_{res_i} are the corresponding mean pixel values; $\sigma_{\text{ref}_i}^2$ and $\sigma_{\text{res}_i}^2$ are the variances; and $\sigma_{\text{ref}_i,\text{res}_i}$ is the covariance between the two bands. $\text{mse}(\cdot,\cdot)$ denotes the mean squared error, and MAX is the maximum possible pixel value. Constants $C_1 = (0.01 \cdot L)^2$ and $C_2 = (0.03 \cdot L)^2$ are used to stabilize the SSIM computation, where L is the dynamic range of pixel values.

3) Implementation Details: For both simulated and real-world experiments, we partition the 3-D HSI cubes from the four aforementioned datasets into training and test sets. Each sample has a spatial size of 256×256 pixels and 28 spectral channels. In total, 1280 images are used for training and 150 for testing, with no overlapping samples between the two sets. The SUT block is trained on the CAVE dataset in a self-supervised manner [34]. The proposed method is compared against several state-of-the-art methods, including BM4D [15], Enhanced3DTV [53], LRTDTV [54], R-PCA [55], 3DCTV-RPCA [56], HLRTF [59], HySuDeep [26], SMDS-Net [28], and HIR-Diff [33]. Parameters for these methods follow those reported in their original papers, with slight fine-tuning to optimize performance. All experiments are conducted on an Intel[6] Core[7] i7-13700KF CPU and MATLAB 2022b. The SUT is evaluated on a single NVIDIA RTX 4090D GPU (24 GB), whereas the diffusion model employs two NVIDIA RTX 4090D GPUs for parallel computing. The simulated Gaussian noise is categorized into two different cases.

4) Abundance-Based Diffusion Model Training Setting: Our model is a conditional DDPM based on ϵ-prediction with mean-squared error (mse) loss. For the noise schedule, we employ the cosine schedule following improved DDPM [60] and set the total diffusion steps to $T = 100$ for both training and inference. The noise parameters are $\beta_{\text{start}} = 1 \times 10^{-6}$ and $\beta_{\text{end}} = 1 \times 10^{-2}$. At each time step, the first eigenimage is incorporated as interpretable guidance to the diffusion model. For training, we use the Adam optimizer ($\beta_1 = 0.9$ and $\beta_2 = 0.999$) with a learning rate of 1×10^{-4} and a batch size of 4 for 300k iterations. To enhance training stability, we apply an exponential moving average (EMA) for the model weights with a decay rate of 0.9999.

B. Simulated HSI Denoising

The proposed model is initially trained under two levels of Gaussian noise.

Case 1: Zero-mean Gaussian noise is added to all spectral bands, with the standard deviation in each band randomly sampled from the range [0, 0.2].

Case 2: Similar to Case 1, but the standard deviation is randomly sampled from the range [0, 0.3].

1) Quantitative Comparison: We present in Table I the performance metrics evaluation of the proposed method and all the comparative methods in the synthetic experiments, where the **bold** represents the best values and the underline denotes the suboptimal values. As shown in Table I, our method achieves the highest PSNR, SSIM, and the lowest ERGAS in most synthetic cases, particularly in terms of SSIM. This superior performance is attributed to DDPM's ability to effectively model contextual relationships, spectral distribution patterns, and material mixing characteristics. However, we observe a slight performance drop on the ICVL dataset, possibly due to its distinct spectral characteristics or scene complexity compared to other datasets. Interestingly, ULADiff attains the best results on the Harvard University dataset. This may be attributed to the relatively regular spatial–spectral structure of the Harvard University dataset that better conforms to the model's learned distribution.

2) Visual Comparison: To facilitate visual comparison, Figs. 6 and 7 illustrate the noise removal and detail preservation ability of the evaluated methods. The region of interest (red rectangular area) is zoomed in and displayed at the

[5]http://vision.seas.harvard.edu/hypers

[6]Registered trademark.

[7]Trademarked.

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TABLE I

PERFORMANCE COMPARISON OF ALL METHODS UNDER DIFFERENT DATASETS AND NOISE CONDITIONS

Dataset	Case	Metric	Observed	BM4D (TIP 12)	LRTDTV (JSTAR 17)	R-PCA (JACM 11)	3DCTV-RPCA (TPAMI 23)	Enhanced3DTV (TIP 20)	HLRTF (CVPR 22)	HIR-Diff (CVPR 24)	Ours
feathers (CAVE dataset[2])	Case 1	PSNR ↑	16.4151	34.9482	30.7212	25.3661	32.8656	36.3404	36.1677	34.9429	**37.0283**
		SSIM ↑	0.1626	0.9079	0.8530	0.8610	0.8250	0.9526	0.8614	0.8775	**0.9750**
		ERGAS ↓	865.5422	72.6902	115.7306	216.3355	90.2653	67.8515	63.4281	127.8621	**59.0325**
	Case 2	PSNR ↑	14.6542	33.8463	29.5085	24.2361	31.9464	35.3982	34.3347	33.9429	**36.5020**
		SSIM ↑	0.1340	0.8878	0.7881	0.6026	0.7886	0.9610	0.8191	0.8301	**0.9684**
		ERGAS ↓	1060.1	82.7646	146.0315	246.2409	99.9124	72.0211	80.6949	139.4318	**69.5336**
KAIST_scene25 (KAIST dataset[3])	Case 1	PSNR ↑	16.4151	31.5481	28.3507	22.5581	29.3770	32.2951	31.7744	30.5526	**33.7567**
		SSIM ↑	0.1938	0.8326	0.7965	0.5304	0.7320	0.8616	0.7861	0.7956	**0.8887**
		ERGAS ↓	755.9496	94.2229	132.7013	240.0888	107.3587	97.7970	115.6781	133.9878	**89.8912**
	Case 2	PSNR ↑	14.6542	30.7427	27.5320	21.4049	28.6780	30.6228	29.4857	29.4430	**31.4287**
		SSIM ↑	0.1617	0.8150	0.7453	0.4777	0.6986	0.8566	0.6892	0.7735	**0.8664**
		ERGAS ↓	925.8392	105.3073	160.8515	271.2192	117.4637	**108.1702**	165.8449	148.3582	124.5195
gavyam_0823-0930 (ICVL dataset[4])	Case 1	PSNR ↑	16.4151	30.5813	27.1814	25.7717	31.8806	**32.2746**	29.9847	31.6524	31.9287
		SSIM ↑	0.2117	0.8107	0.7244	0.7400	0.8349	0.8620	0.7281	0.7845	**0.8674**
		ERGAS ↓	1011.2	135.7735	226.8262	200.3412	154.6774	158.1673	146.1996	132.6292	**128.5195**
	Case 2	PSNR ↑	14.6542	29.5570	26.4507	24.6750	30.8258	**31.0597**	27.4119	30.9315	30.9077
		SSIM ↑	0.1742	0.7803	0.6804	0.6930	0.8007	0.8340	0.6075	0.8230	**0.8492**
		ERGAS ↓	1238.5	152.6956	249.8677	227.0193	178.1130	164.5718	313.9522	**142.5432**	146.7308
img1_Harvard (Harvard University dataset[5])	Case 1	PSNR ↑	16.4151	32.2615	29.2260	28.7898	33.4134	33.4610	34.4962	31.7592	**37.1323**
		SSIM ↑	0.1681	0.7951	0.7056	0.7387	0.8235	0.8315	0.8367	0.8188	**0.9341**
		ERGAS ↓	1617.0	157.5545	283.6896	192.6068	197.1345	226.0446	91.3908	130.6512	**67.2778**
	Case 2	PSNR ↑	14.6542	31.3763	28.2926	27.6981	32.3884	32.8447	30.8057	30.8649	**36.8066**
		SSIM ↑	0.1354	0.7688	0.6583	0.6883	0.7900	0.8248	0.6262	0.7867	**0.9469**
		ERGAS ↓	1980.4	177.0674	348.2871	219.3386	228.0938	259.6702	228.4176	147.7609	**123.3956**

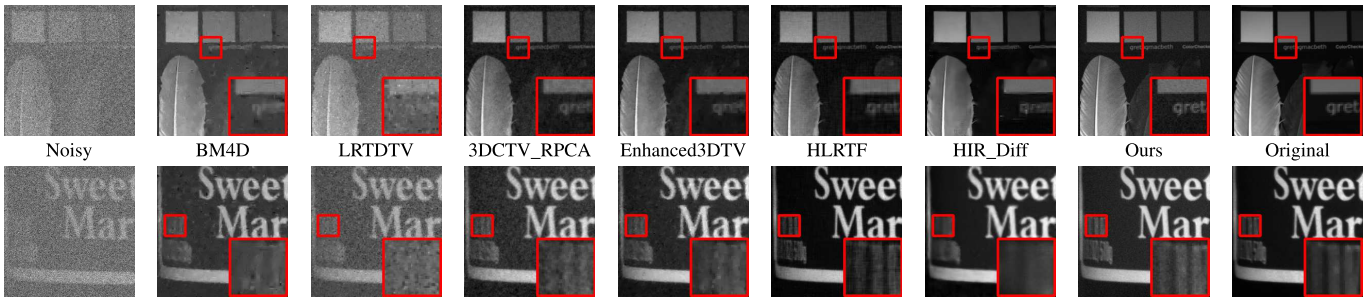


Fig. 6. Visual comparison on CAVE (the second band) and KAIST (the tenth band) dataset by different HSI denoising methods.

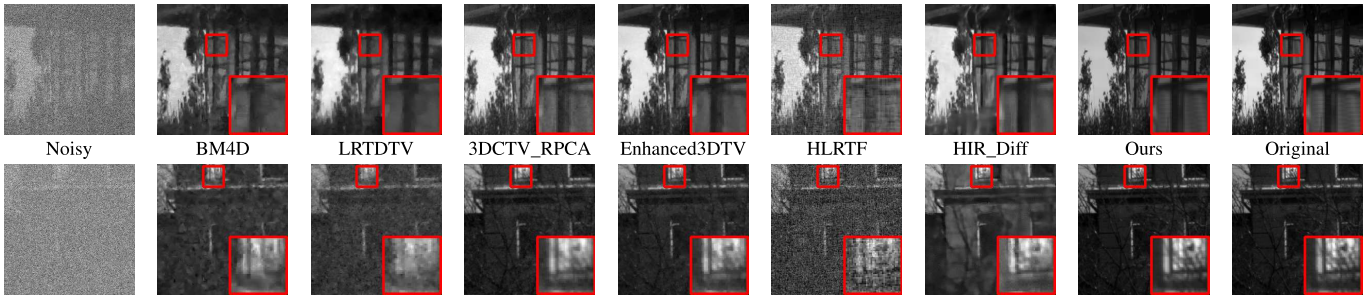


Fig. 7. Visual comparison on ICVL (the fourth band) and Harvard University (the tenth band) dataset by different HSI denoising methods.

bottom right of each image. From Figs. 6 and 7, we find that our method, which leverages the powerful generative ability of DDPM and priors of HSIs, produces visually superior results and demonstrates enhanced detail preservation. Here, BM4D, LRTDTV, and 3DCTV-RPCA fail to completely remove the noise visually. On the other hand, HLRTF introduces slight artifacts. Although Enhanced3DTV and HIR-Diff reduce noise, their ability to recover local details still has room for further improvement. In contrast, the proposed method restores images with higher spectral fidelity and spatial detail, demonstrating its robustness in noise removal. We present results on three of the four datasets, KAIST, ICVL, and Harvard University datasets. As shown in Figs. 8–10, the curve at multiple representative pixels of our method is the closest to that of the original HSI across all bands and typical regions. This confirms the highest spectral fidelity of our method.

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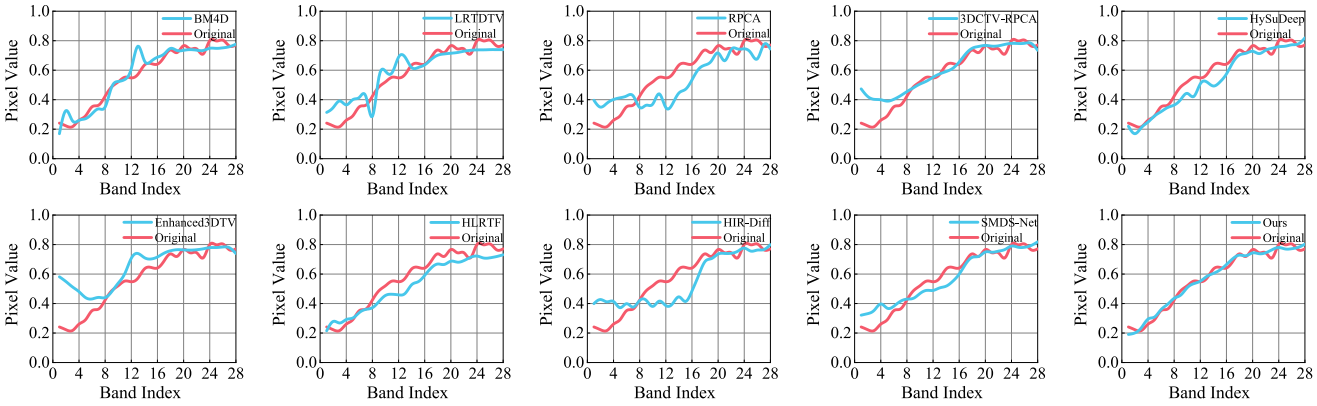


Fig. 8. Spectral curves at a representative pixel (M = 113 and N = 147) in KAIST_scene25.

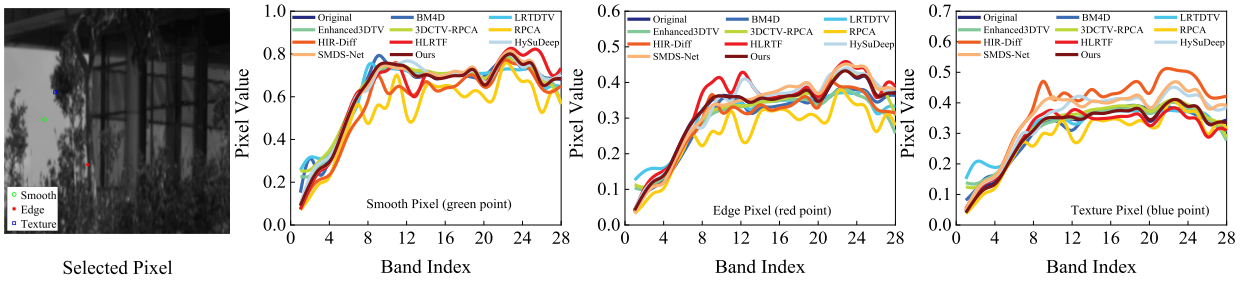


Fig. 9. Spectral curves at representative smooth (M = 46 and N = 127), edge (M = 95 and N = 178), and texture (M = 58 and N = 96) pixels on gavyam_0823-0930 (Case 2).

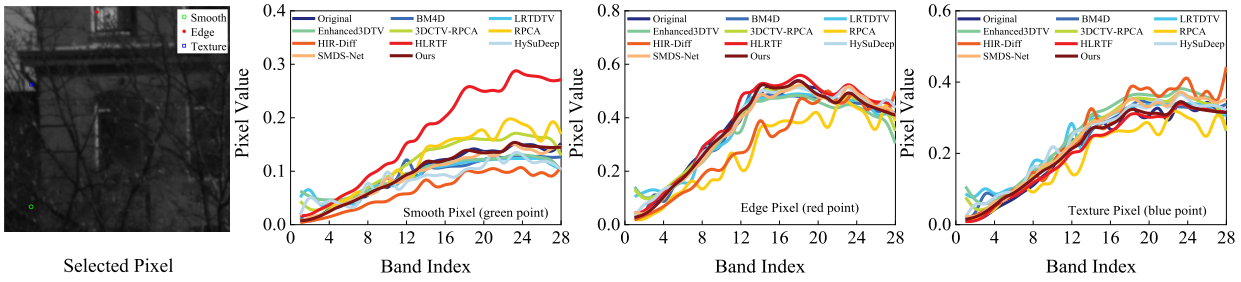


Fig. 10. Spectral curves at representative smooth (M = 31 and N = 227), edge (M = 105 and N = 7), and texture (M = 31 and N = 89) pixels on imga1_08 (Case 2).

C. Real HSI Denoising

The Urban dataset[8] is one of the most widely used HSI benchmarks. It comprises a 307 × 307 image, with each pixel covering approximately 2×2 m^2 surface area. The data cube provides 210 spectral bands spanning 400–2500 nm at 10-nm spectral resolution. Moreover, the Urban dataset includes real-world noise, such as stripes, dead lines, atmospheric interference, water absorption, and other unidentified sources. In addition, the Australia[9] dataset, an EO-1 Hyperion scene dataset, is also used for evaluation. As shown in Fig. 11, we provide a visual comparison of denoising results on both datasets. It is evident that Enhanced3DTV, 3DCTV-RPCA, HIR-Diff, and our method exhibit superior denoising performance. Notably, our model effectively removes complex noise while maintaining high fidelity in texture details, thereby achieving superior results.

D. Training Complexity Analysis

As shown in Table II, the diffusion component dominates the training-time computational cost and memory usage, making it the primary bottleneck. In contrast, the SUT module remains lightweight.

Specifically, the memory overhead of diffusion models mainly arises from: 1) storing large activations and weights for back-propagation [62]; 2) iterative, full-precision noise estimation across multiple time steps [63]; and 3) the feature maps in the diffusion U-Net, which scale with the spatial resolution of the latent features. To mitigate this, existing strategies include latent diffusions [61], quantized diffusion [62], [63], and memory-efficient attention mechanisms [64].

[8]http://www.tec.army.mil/hypercube/
[9]http://remote-sensing.nci.org.au/u39/public/html/index.shtml

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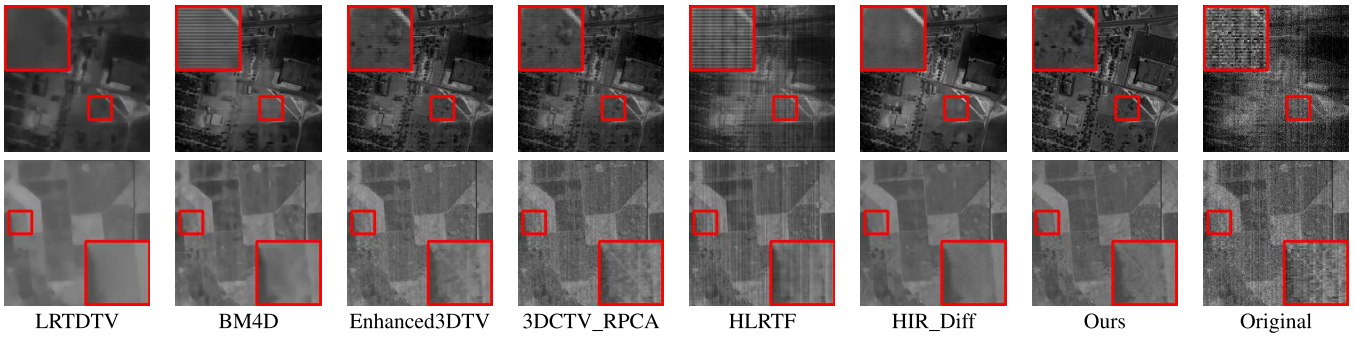


Fig. 11. Restoration results on the 108th band of the real Urban, and the 49th band of the real Australia. Highlighted regions demonstrate that our method better preserves structural details and suppresses noise, particularly in building edges (Urban, the first row) and field boundaries (Australia, the second row).

TABLE II
COMPUTATIONAL COST AND RESOURCE USAGE DURING TRAINING FOR THE SUT MODULE AND THE LATENT DIFFUSION COMPONENT

Component	GFLOPs	Params (M)	GPU Memory (GB / device)	Time / Epoch (s)	Notes
SUT module	16.66	1.27	2.47	32.62	Lightweight component
Latent diffusion	717.37	391.05	21.70	415.93	Main computational bottleneck

Note: All measurements are reported for a global batch size of 4 at a resolution of $256 \times 256 \times 28$, where GFLOPs are forward-only. The SUT module is evaluated on a single RTX 4090D (24 GB), whereas the diffusion model is trained using PyTorch DDP across two RTX 4090D GPUs. Peak memory is obtained via `torch.cuda.max_memory_allocated`, and per-epoch time is averaged over five runs with `torch.cuda.synchronize()`.

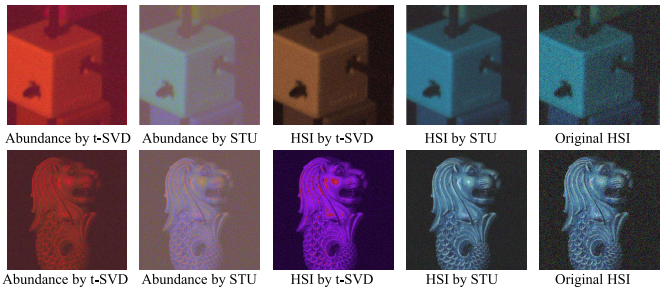


Fig. 12. Comparison of unmixing results using t-SVD and SUT. Columns 1 and 2: abundance maps (R:1, G:2, and B:3), Columns 3 and 4: reconstructed HSIs (R:5, G:15, and B:25), and Column 5: original HSIs (R:5, G:15, and B:25) [34].

E. Ablation Study

To evaluate the contributions of the proposed method, we conduct a series of ablation studies on the CAVE and KAIST datasets. All ablation experiments are performed on a single NVIDIA RTX 4090D GPU.

1) Spectral Unmixing Transformer: To evaluate the effectiveness of the SUT module [34], we perform visual analyses of the decomposition network and compare its performance against other unmixing methods. It is important to note that spectral unmixing is inherently an ill-posed problem, as it lacks a unique optimal solution. A key consideration in decomposition is maintaining the consistency of endmember representations under varying noise conditions. For comparison, we also evaluate SVD and DL-based techniques, as illustrated in Fig. 12. The HSI reconstructed using tensor SVD (t-SVD) exhibits noticeable deviations from the original data. In contrast, the SUT module achieves better reconstruction by effectively integrating abundance maps with their corresponding endmembers. In addition, traditional methods such as principal component analysis (PCA) and virtual dimensionality (VD) are widely applied for dimensionality reduction and reconstruction. However, they often neglect the physical interpretability and intrinsic spectral correlations of HSIs. For instance, PCA is restricted to linear transformations. In contrast, the SUT not only reduces dimensionality but also effectively overcomes these limitations.

2) Reconstruction Module: To assess the contribution of the reconstruction module, we perform an ablation study by comparing the model performance with and without the spatial–spectral consistency priors. As shown in Table III, removing these priors (w/o priors) leads to a noticeable degradation in denoising performance across all metrics. Specifically, under the Gaussian noise setting, the model without priors yields lower PSNR and SSIM, and higher ERGAS values. This confirms that the proposed priors play a crucial role in preserving spatial–spectral structures and enhancing overall reconstruction quality.

3) Comparison of Inference Time: To further assess the efficiency of our method, we compare its inference time to that of a band-wise denoising strategy using a pretrained three-band diffusion model.[1] Given that test data contain 28 spectral bands, directly applying the three-band model requires additional preprocessing to match the input dimensionality. To this end, we design two strategies: 1) T_{27}: we discard the last spectral band, grouping the remaining 27 bands into nine triplets; and 2) T_{30}: two all-one channels are padded to reach 30 bands, allowing for ten triplets to be processed. Each strategy requires multiple samples, leading to significant computational overhead. In contrast, our method performs inference over the entire HSI in a single inference, effectively avoiding redundant computation. All methods are evaluated under the same experimental setting, with the number of sampling time steps fixed at 100 and a cosine noise schedule

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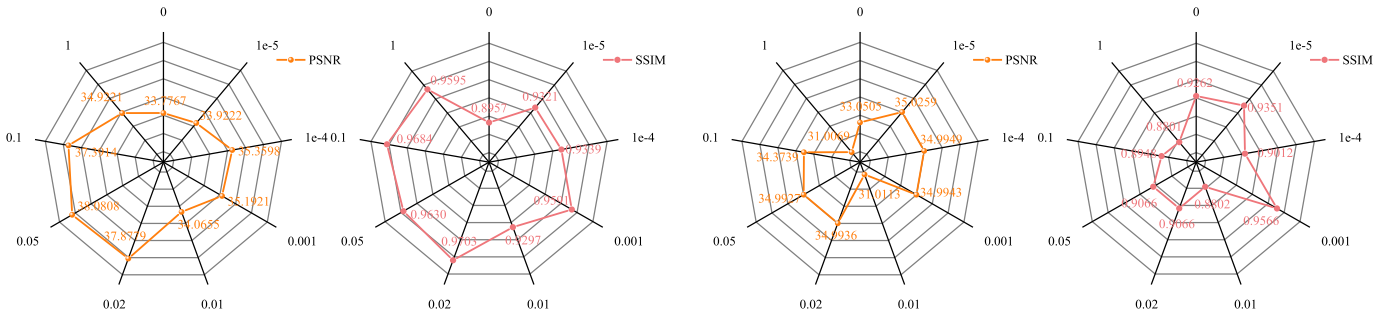


Fig. 13. (a) Impact of λ_2 on the evaluation metrics when λ_1 is fixed at 0.001 and (b) impact of λ_1 on the evaluation metrics when λ_2 is fixed at 0.05, evaluated on the CAVE dataset.

TABLE III

QUANTITATIVE COMPARISON ON CAVE AND KAIST DATASETS (CASES 1 AND 2). W/PRIORS DENOTES ULADIFF WITH SPATIAL–SPECTRAL PRIORS AND W/O PRIORS INDICATES THE VARIANT WITHOUT PRIORS

Dataset	Case	Metric	*w/ priors*	*w/o priors*
feathers (CAVE dataset[2])	Case 1	PSNR ↑	37.0283	35.7767
		SSIM ↑	0.9750	0.9457
		ERGAS ↓	59.0325	69.1098
	Case 2	PSNR ↑	36.5020	35.5324
		SSIM ↑	0.9684	0.9083
		ERGAS ↓	69.5336	75.5354
KAIST_scene25 (KAIST dataset[3])	Case 1	PSNR ↑	33.7567	32.0760
		SSIM ↑	0.8887	0.8366
		ERGAS ↓	89.8912	106.3465
	Case 2	PSNR ↑	31.4287	30.5363
		SSIM ↑	0.8664	0.8408
		ERGAS ↓	124.5195	136.0098

TABLE IV

COMPARISON OF INFERENCE TIME (S/IMG) BETWEEN BAND-WISE DDPM AND ULADIFF ON BALLOONS_MS AND KAIST_SCENE04

Dataset	Strategy	Inference time (s/img)
balloons_ms (CAVE)	Band-wise DDPM (T_{27})	279.7325
	Band-wise DDPM (T_{30})	312.5018
	ULADiff (ours)	**33.7576**
KAIST_scene04 (KAIST)	Band-wise DDPM (T_{27})	281.6970
	Band-wise DDPM (T_{30})	313.1352
	ULADiff (ours)	**34.7885**

employed to ensure a fair comparison. As summarized in Table IV, our method not only achieves superior denoising performance but also substantially reduces inference time. This highlights the practical advantage of our ULADiff for real-time applications.

4) Parameter Sensitivity Analysis: To investigate the impact of the hyperparameters λ_1 and λ_2, we conduct a detailed sensitivity analysis on the CAVE dataset. As illustrated in Fig. 13, when λ_1 is fixed at 0.001, PSNR steadily increases with larger λ_2 values and peaks at $\lambda_2 = 0.05$. Beyond this point, further increasing λ_2 leads to a notable performance decline. This indicates that a moderate λ_2 value helps enforce the spatial–spectral consistency and improves reconstruction quality, whereas an overly large weight may hinder denoising. Based on these observations, we empirically select $\lambda_1 = 0.001$ and $\lambda_2 = 0.05$ as the optimal hyperparameter settings for subsequent experiments.

TABLE V

QUANTITATIVE COMPARISON OF THE PROPOSED METHOD WITH HYSUDEEP AND SMDS-NET ON VARIOUS HSI DATASETS. "OBSERVED" INDICATES THE NOISY INPUT, ↑ DENOTES HIGHER IS BETTER, AND ↓ DENOTES LOWER IS BETTER

Dataset	**Case**	**Metric**	**Observed**	**HySuDeep**	**SMDS-Net**	**Ours**
feathers (CAVE dataset[2])	Case 1	PSNR ↑	16.4151	34.1294	35.2145	**37.0283**
		SSIM ↑	0.1626	0.8684	0.9233	**0.9750**
		ERGAS ↓	865.5422	93.4758	128.3544	**59.0325**
	Case 2	PSNR ↑	14.6542	32.9673	33.8167	**36.5020**
		SSIM ↑	0.1340	0.8485	0.8991	**0.9684**
		ERGAS ↓	1060.1	110.3997	135.9929	**69.5336**
KAIST_scene25 (KAIST dataset[3])	Case 1	PSNR ↑	16.4151	30.3570	31.0474	**33.7567**
		SSIM ↑	0.1938	0.8662	0.8729	**0.8887**
		ERGAS ↓	755.9496	97.1798	82.0264	**89.8912**
	Case 2	PSNR ↑	14.6542	29.6560	27.4806	**31.4287**
		SSIM ↑	0.1617	0.8466	0.7902	**0.8664**
		ERGAS ↓	925.8392	113.5096	127.4764	**124.5195**

F. Discussions

1) Comparison With PnP and DUN Methods: To further evaluate our method, we compare it with hybrid paradigms integrating the strengths of model-based and deep-learning methods. Specifically, we employ HySuDeep [26] for PnP and SMDS-Net [28] for DUN. For fairness, HySuDeep is run using the official implementation, whereas SMDS-Net is retrained on the ICVL dataset under our noise setting. The corresponding quantitative results are summarized in Table V.

As shown in Table V, ULADiff outperforms HySuDeep and SMDS-Net in PSNR and SSIM across all datasets and noise cases. HySuDeep, however, is designed for mixed noise scenarios, explaining its suboptimal performance under purely Gaussian noise. While SMDS-Net offers interpretability via iterative unfolding, its patch-based processing requires aggregation that may introduce artifacts.

2) Illustration of Physical Interpretability: In our work, "physical interpretability" means that the SUT decomposes an HSI into endmembers and abundance maps that adhere to the LMM, rather than opaque features from autoencoders [47].

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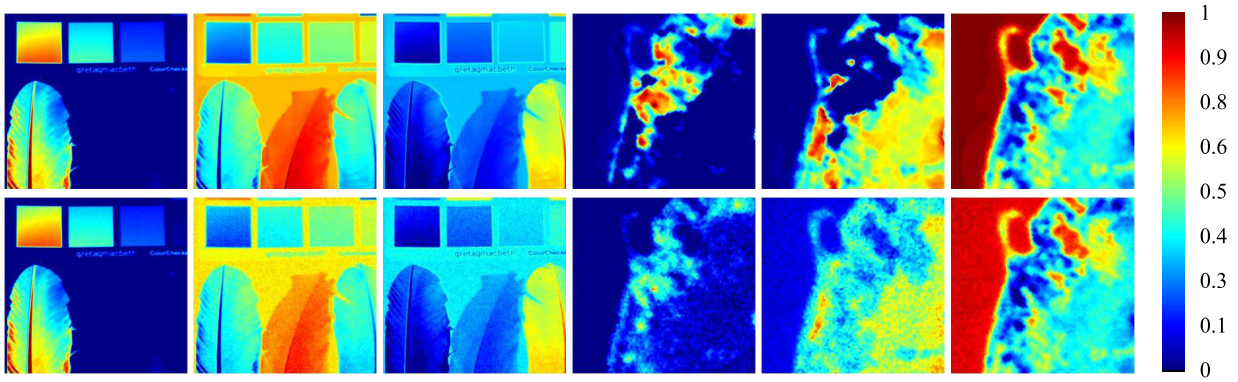


Fig. 14. Comparison of abundance maps for the feathers_ms (CAVE dataset) and the Samson dataset. (Top row) Reference abundance maps and (bottom row) estimated abundance maps from noisy input processed by the SUT module.

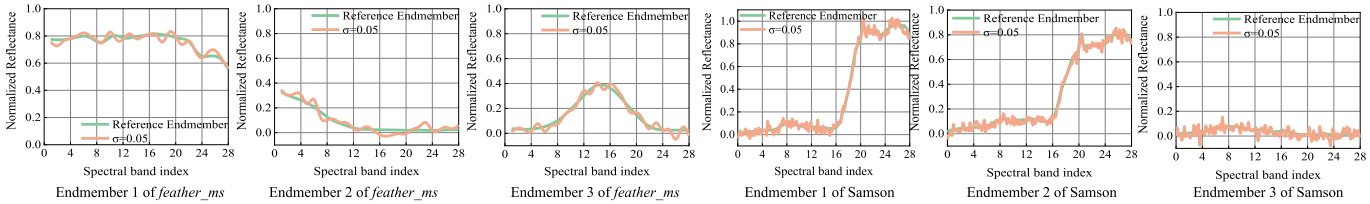


Fig. 15. Endmember signatures for feathers_ms (CAVE dataset) and the Samson dataset. Orange: estimates from noisy inputs processed by the SUT module, and green: reference endmembers. The close overlap indicates that the learned endmembers are relatively invariant with spectral variability.

Accordingly, our diffusion model operates in the abundance latent space. To validate this, we provide two forms of evidence.

1) The endmembers extracted by the SUT exhibit relative invariance under spectral variability. Furthermore, the abundance maps satisfy the ANC and ASC.
2) We provide a visual comparison of the endmembers and abundance maps generated by the SUT on the CAVE and Samson[10] datasets under different noise cases.

To validate the first claim, we vary the Gaussian noise level (e.g., $\sigma \in \{0.03, 0.10\}$) using a fixed random seed. After matching, the spectral angle mapper (SAM) between matched endmembers remains low. Meanwhile, the estimated abundance maps satisfy the ASC and ANC. Specifically, on *feathers_ms*, after the SUT, negative entries drop to 0% with a sum-to-one residual of $\approx 1.3 \times 10^{-8}$. Given $\tau = 0.15$, the average active endmembers per pixel is 2.43, and the pure-pixel ratio (>0.9) is 0.03%. These results affirm the physical interpretability of the SUT.

Visual comparisons in Figs. 14 and 15 demonstrate that the abundance maps estimated by SUT preserve spatial structures and material boundaries accurately, while the extracted endmember spectra remain consistent under different noise levels.

3) Limitations: Artifacts and Color Deviations: Although ULADiff improves denoising, we sometimes observe artifacts and color shifts in some datasets. To explore the potential cause, we analyze the distributions of latent features. Fig. 16 reveals the misalignment between the generated latent features and the ground truth (GT), which is consistent with previous studies [65], [66]. A future direction is to use color patterns as an additional condition to guide the diffusion sample process.

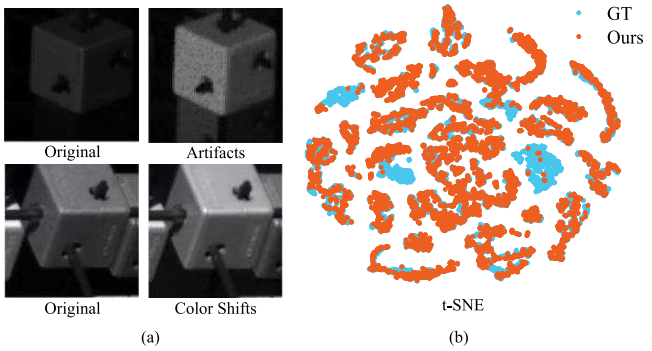


Fig. 16. Limitation analysis. (a) Examples of artifacts and color shifts. (b) t-SNE visualization of feature distributions for our generated images and GT.

4) Application in Downstream Tasks: To demonstrate the practical utility of our denoising method, we evaluate its performance on the downstream task of HSI classification using the Pavia University dataset.[11] The original image is degraded by Gaussian noise (standard deviation $\sigma = 0.2$). The noisy image is then denoised by several methods, including BM4D, RPCA, LRTDTV, 3DCTV-RPCA, Enhanced3DTV, HIR-Diff, HLRTF, and our method. For evaluation, a 3DCNN classifier [67] is exclusively trained on the original image. The fixed 3DCNN is then applied to all denoised outputs to obtain classification maps, with the overall accuracy (OA) as the evaluation metric. We use 3% and 97% of the samples for training and testing [68], respectively.

[10] http://opticks.org/confluence/display/opticks/Sample+Data

[11] https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

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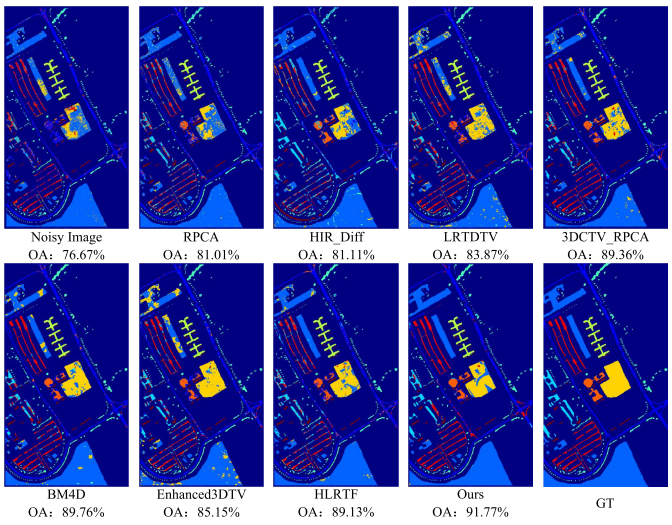


Fig. 17. HSI classification results using a 3DCNN classifier [67] on the Pavia University dataset. A comparison of the proposed and state-of-the-art denoising methods.

Quantitative results and the corresponding visualizations are provided in Fig. 17. As shown in Fig. 17, our method achieves the best performance on easily confused categories (e.g., Asphalt, Bitumen, and Bricks). This demonstrates its ability to preserve spatial–spectral features essential for distinguishing dark pavements from brick surfaces.

V. Conclusion

In this article, we propose ULADiff, a physically interpretable diffusion framework for HSI denoising. By integrating spectral unmixing with abundance-guided diffusion, ULADiff incorporates physical priors into DDPMs. Specifically, ULADiff performs restoration by reconstructing the HSI using the endmembers estimated by the SUT module and the abundance maps generated by the DDPM with our designed guidance. Finally, we impose a spatial–spectral consistency prior to restoring HSIs with higher fidelity. Experimental results demonstrate that ULADiff achieves both visual and quantitative improvements while significantly reducing inference time.

While ULADiff proves effective for HSI denoising, several improvements can be explored. First, we can extend the method to other HSI restoration tasks, such as super-resolution, inpainting, and reconstruction. Second, to address the color shifts observed in some generated results, a dedicated color correction module could be designed to mitigate color shifts in the diffusion output.

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